05. Train-Test Split & Validation Strategy

Separate the original training protocol from the evaluation notebook run.

Project: Automated Mobile-Technology Brand Sentiment and Headline Generator

Source archive: NLP-Sentiment-Classification-of-Mobile-brands-main.zip | Notebook: main.ipynb | Supporting report and presentation included

Original report split

The project report says articles and tweets were randomly split in a stratified manner into 3,000 training, 600 validation and 400 test examples each. This implies six separate partitions: article train/validation/test and tweet train/validation/test. The validation set supports model selection and early stopping; the test set should be held for final reporting.

Notebook role

The supplied notebook reads evaluation_data.xlsx and runs inference using pretrained model artifacts. It is not a model-training notebook for all pipeline stages. The training data and model-building modules are not included in the ZIP, so the training split cannot be independently reconstructed here.

Validation by task

- Theme classification: precision, recall, F1 and accuracy, separately for article and tweet domains; weighted F1 is mentioned in the report.
- Brand identification: entity-level accuracy/precision/recall, including aliases, false matches and missed mentions.
- Sentiment: evaluate per-brand labels with precision, recall and F1; avoid scoring the whole document as one sentiment.
- Headline generation: ROUGE, BLEU and embedding-based similarity against reference headlines; also human factuality and readability review.
- Runtime: measure preprocessing, translation, model inference and end-to-end latency separately.

Leakage and split risks

- Ensure duplicate or near-duplicate articles/tweets do not cross train/validation/test partitions.
- If data is collected over time, use a temporal test set to measure future robustness.
- Keep translations and augmentations derived from a source record in the same split as that record.
- Do not tune headline generation or sentiment thresholds on the test set.
- Evaluate language slices and single-brand versus multi-brand cases independently.

Interview response

> “The project report describes stratified 3,000/600/400 splits for articles and tweets. The notebook I inspected is an inference/evaluation run over a separate sample; it loads pretrained vectorizers, classifiers and neural checkpoints. I would keep the final test split untouched and report metrics separately for articles, tweets and language groups.”